DS351L - Data Visualization Lab

LAB07 *2nd April 2026*

Task: Calculated Fields

- Create two calculated fields using your dataset.
- Use those fields in a meaningful visualization of your choice.

Task: Aggregation Basics

Compare different aggregation methods.

1. Create a bar chart with a categorical field (e.g., Category, Region, Product Type) on Columns.
2. Place a numeric measure (e.g., Sales, Price, Quantity) on Rows.
3. Observe the default aggregation (usually SUM).
4. Change aggregation to AVG, MIN, and MAX.

Task: Time Series Visualization

Visualize trends over time.

1. Use a Date field → Columns (set to Month or Year).
2. Place a numeric measure (e.g., Sales, Price, Units) → Rows.
3. Add a categorical field (e.g., Region/Category) → Color.

Task: Trend Line Analysis

Add statistical modeling to a time series.

1. Create a line chart or use an already created one.
2. Go to Analytics Pane → Add Trend Line (Linear).
3. Compare trends across categories or regions.

Task: Dashboard

Combine insights into a single interactive view.

1. Create a dashboard with all the sheets you have created.
2. Add filters (e.g., Category, Region, Year) → apply to all sheets.
3. Add titles and arrange the layout cleanly.

Task: Dashboard Creation with Interactive Filters

Combine multiple visualizations into a single interactive dashboard using a dataset of your choice.

1. **Play Around with Your Data**:
   - Use a dataset of your choice. Explore and experiment with different visualizations to uncover insights.
   - Adjust the visualizations to make sure they are clear and informative, ensuring the relationships in the data are easy to interpret.
   - Experiment with different types of charts (e.g., bar charts, line charts, stacked bar charts) and styles to find the best representation for your data.
2. **Create a Dashboard**:
   - Create a new dashboard and add your visualizations (minimum 3) to it. These should represent various aspects of the data that you've explored.
3. **Add Interactive Filters**:
   - Incorporate filters into your dashboard to allow for dynamic interaction with the data.
   - Ensure that users can apply these filters and see how the data changes based on their selections.
4. **Enhance Visual Appeal**:
   - Use appealing colors to distinguish different categories or measures.
   - Ensure the dashboard is visually cohesive and easy to read, with clear labels and legends.
   - Make sure the interactions between filters and charts are smooth and intuitive.